
Speaker notes

The presentation - docs/slides.pptx

9 slides, about 3 minutes

What to say on each slide, what to point at, and short answers to the questions each slide may bring up.

DA 2009 - Data Collection Methods II

Assignment 2 (Group Project)

University of Colombo

25ada072 - Pasindu | 25ada073 - Sasini | 25ada141 - Salaama | 24ada076 - Nuhan

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How the presentation works

The viva is 8 minutes for the whole team: about 3 minutes of presentation, then 5 minutes of questions on what each member did. Any number of us can present.

- **Slides 1 and 2** - Nuhan opens.
- **Slides 3 to 9** - any member who wants to present. The script is written so that anyone can read it.

Slide	Starts at	Who	Words
1. Collecting data from the Colombo Stock Exchange	0:00	Nuhan	38
2. What we collect	0:16	Nuhan	53
3. The surprise: the page is built by JavaScript	0:38	Any member	51
4. The answer: read the data the page itself uses	0:59	Any member	55
5. Ethics: every request goes through one function	1:22	Any member	53
6. Announcements: finding and reading the PDFs	1:43	Any member	52
7. OCR: reading a scanned page	2:05	Any member	55
8. Clean, check, chart and save	2:28	Any member	61
9. Who did what	2:53	Any member	41
End	3:10		

The times assume a calm pace of about 145 words a minute. Speaking a little faster than that is fine; rushing is not.

Sharing the slides between volunteers

Volunteers	Suggested split after Nuhan's opening
1	Slides 3 to 9
2	Slides 3 to 5, then 6 to 9
3	Slides 3 and 4, then 5 and 6, then 7 to 9

When you hand over, keep it short - for example:

"Now Sasini will show how we found the documents."
"Over to Salaama for the scanned page."

Remember: the words on each slide are only reminders. Look at the audience, not the screen, and point at the part of the slide you are talking about.

Before you start

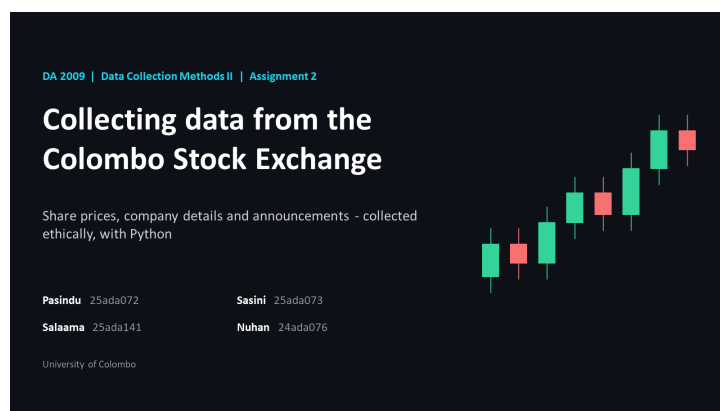
1. Open docs/slides.pptx in PowerPoint on the laptop you will present from.
2. On the **Slide Show** tab, tick **Use Presenter View**, then press **F5**. The audience sees the slide; your screen also shows these notes and a timer.

3. Move with the arrow keys or a clicker. **Esc** ends the show.
4. Practise out loud at least twice with a timer - once alone, and once with the whole team handing over to each other.
5. Keep the app ready in the background (double-click 2_START_The_App.bat), in case the examiner asks to see something running. The live demo notes list what to show for each member.

Be careful: the numbers on the slides - 284 companies, 96 rose, 113 fell - are from one trading day, 25 September 2026. If you are asked, say that they change every day, and the app always shows the latest.

The script, slide by slide

Slide 1 - Collecting data from the Colombo Stock Exchange



Time	0:00 to 0:16 (about 16 seconds)
Who	24ada076 - Nuhan

Say

Good morning. I'm Nuhan, and this is our DA 2009 group project: collecting data from the Colombo Stock Exchange website. Our group is Pasindu, Sasini, Salaama and me.

I'll open, and my teammates will explain how it works.

Point to

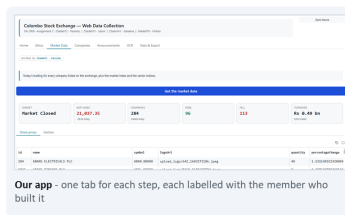
- The four names on the slide as you say them.

Slide 2 - What we collect

What we collect

Three kinds of public data from www.cse.lk - no logins, nothing personal

- 1 Share prices**
Today's price, change and turnover for every listed company - about 284 of them.
- 2 Company details**
One company at a time: its market value, and its highest and lowest price this year.
- 3 Announcements**
Notices companies publish - many as PDF files, and some of those are scans.



2

Time	0:16 to 0:38 (about 22 seconds)
Who	24ada076 - Nuhan

Say

We collect three kinds of public data: share prices for every listed company, details for each company, and company announcements, many of them as PDF files. No logins, and nothing personal.

It all runs in this app, with one tab for each step. But our first attempt found almost nothing - here's why.

Hand over to the next speaker.

Point to

- The three numbered items.
- The app screenshot when you say 'this app'.

If you are asked

Why the Colombo Stock Exchange?

It was the website our group was given. It is a good test because it has live data, company records and PDF documents - so it needed several different collection methods.

Slide 3 - The surprise: the page is built by JavaScript

The surprise: the page is built by JavaScript

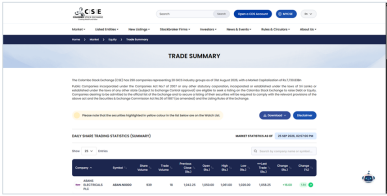
The page arrives almost empty. JavaScript in the browser fills in the prices afterwards.

A plain download
requests + BeautifulSoup

24
characters of text

0 tables, out of 17,898 bytes of HTML. The prices are simply not in the file.

A real browser
Selenium driving Microsoft Edge



4,121 characters and 1 table - but only after waiting 5.3 seconds for one page

Time	0:38 to 0:59 (about 21 seconds)
Who	Any member

Say

We started as the course taught us, with requests and BeautifulSoup. The trade summary page gave just 24 characters of text and no tables.

CSE builds its pages with JavaScript, and requests never runs it. Selenium, a real browser, did see the table - but took over five seconds a page.

Point to

- The red 24 on the left.
- The table in the browser picture on the right.

If you are asked

Why not just use Selenium for everything?

It works, but it is slow - over five seconds a page - and it loads the whole page with its pictures and scripts, which puts more load on CSE. Reading the data addresses directly is faster and lighter.

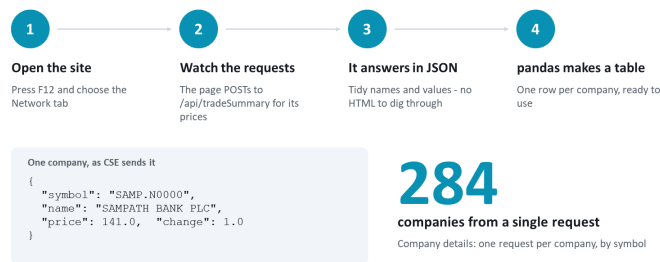
What is a dynamic page?

A page whose content is added by JavaScript after it arrives. A plain download only gets the empty page.

Slide 4 - The answer: read the data the page itself uses

The answer: read the data the page itself uses

Found with the browser's developer tools - no guessing



Time	0:59 to 1:22 (about 23 seconds)
Who	Any member

Say

We pressed F12 and watched the Network tab. The page gets its prices from addresses like `api/tradeSummary`, which answer in JSON.

We ask those addresses directly - most only answer POST - and pandas turns the JSON into a table. One request gives all 284 companies; company details come one at a time, by symbol.

Point to

- The four steps, left to right.
- The JSON record, then the big 284.

If you are asked

Is it allowed to use those addresses?

They are public - the page itself uses them for every visitor, with no login - and `robots.txt` does not block them. We still wait 1.5 seconds between requests and say who we are.

What is the difference between GET and POST?

GET just asks for something. POST sends information along with the request, like a filled-in form. Most CSE addresses only answer POST.

Slide 5 - Ethics: every request goes through one function

Ethics: every request goes through one function

The rules are written once, so no part of the project can skip them

1

Check robots.txt
CSE blocks one folder, /cgi-bin/ - we never go there

2

Wait 1.5 seconds
Between every request, even though CSE does not ask us to

3

Say who we are
Our university, course and an email address in the User-Agent

4

Log it
Every request is written to logs/request_log.csv

5

Keep a copy
So we never ask CSE for the same thing twice

CSE's robots.txt

User-agent: *
Disallow: /cgi-bin/
Sitemap: https://www.cse.lk/sitemap.xml

• Public data only

• Nothing personal is collected

• CSE is credited inside every file we save

5

Time	1:22 to 1:43 (about 22 seconds)
Who	Any member

Say

Ethics is a main part of the marks, so every request goes through one function.

It checks robots.txt - CSE only blocks cgi-bin, which we never use. It waits 1.5 seconds, though CSE doesn't ask us to, says who we are, logs every request, and keeps a copy so we never ask twice.

Point to

- The five numbered steps.
- The Disallow: /cgi-bin/ line in the robots.txt box.

If you are asked

Why wait 1.5 seconds if robots.txt doesn't ask for a delay?

Being allowed to go fast is not a reason to. Sending requests as fast as possible would put load on their server for no benefit to us.

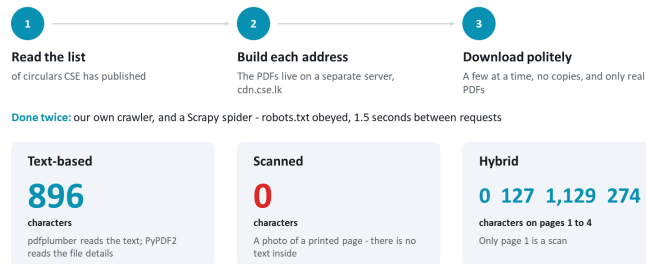
Can you show the requests you sent?

Yes - every one is in logs/request_log.csv, and the Ethics tab of the app shows it.

Slide 6 - Announcements: finding and reading the PDFs

Announcements: finding and reading the PDFs

A crawler finds the documents, then we read the text out of each one



6

Time	1:43 to 2:05 (about 22 seconds)
Who	Any member

Say

Our crawler reads the list of circulars, finds each PDF on CSE's file server, and downloads a few at a time, skipping copies. A Scrapy spider does the same crawl, also obeying robots.txt.

pdfplumber reads the text, PyPDF2 the file details - but one PDF was a scan, with no text inside.

Point to

- The three steps of the crawler.
- The red 0 on the Scanned card.

If you are asked

What is the difference between scraping and crawling?

Scraping takes data from a page you already know about. Crawling finds pages or files by following links. We did not know which documents existed, so we crawled for them.

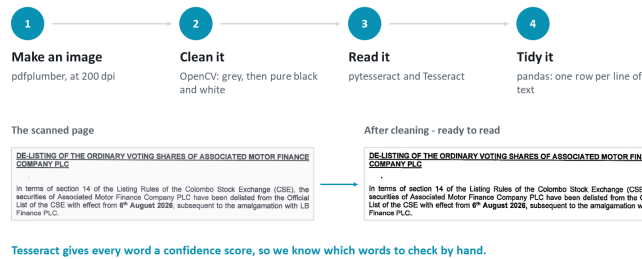
How do you know a download is really a PDF?

Every PDF file starts with the characters %PDF. The crawler checks the first four bytes before saving anything.

Slide 7 - OCR: reading a scanned page

OCR: reading a scanned page

The four steps of the OCR pipeline, one tool for each



Time	2:05 to 2:28 (about 23 seconds)
Who	Any member

Say

For scans we use OCR, in the lecture's four steps. pdfplumber makes an image, OpenCV turns it grey and then pure black and white - you can see the difference here - pytesseract reads the letters, and pandas tidies the result.

Every word gets a confidence score, so we know which to check by hand.

Point to

- The four steps.
- The two pictures - before, and after cleaning.

If you are asked

Why make the image black and white first?

Tesseract reads best with plain black letters on a white background. We use adaptive thresholding, which picks a separate cut-off for each small area, so uneven lighting does not spoil it.

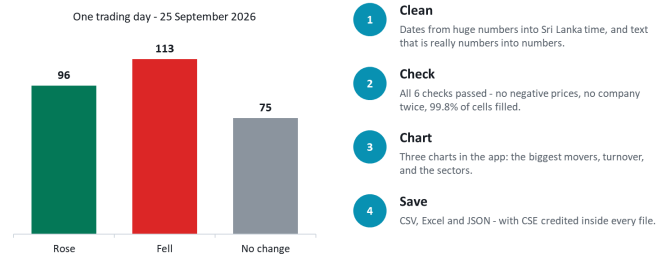
Is OCR always right?

No. Letters like O and 0, or I and 1, get mixed up, and logos confuse it. That is why we show the least confident words, so a person can check them.

Slide 8 - Clean, check, chart and save

Clean, check, chart and save

Data from a website is never ready to use straight away



Time	2:28 to 2:53 (about 25 seconds)
Who	Any member

Say

Website data is never ready to use. We turn dates from huge numbers into Sri Lanka time, and check the values make sense - all six checks passed.

The app draws three charts - movers, turnover and sectors. On this day 96 companies rose, 113 fell and 75 didn't move. We save CSV, Excel and JSON, crediting CSE in every file.

Point to

- The chart as you say the three numbers.
- The four steps on the right.

If you are asked

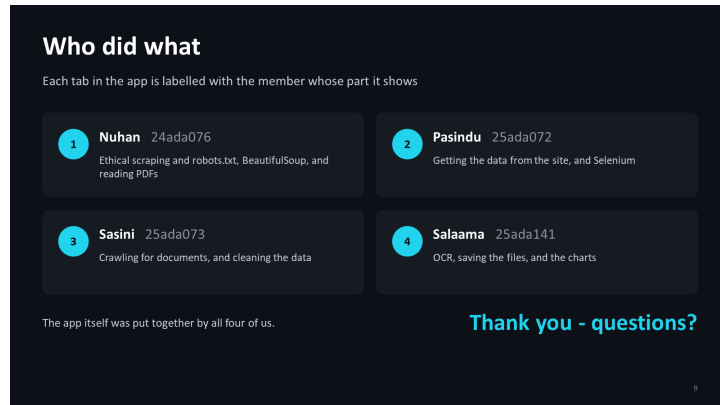
What is the difference between cleaning and checking?

Cleaning fixes the shape - dates, number types, spaces, empty columns. Checking asks whether the values are believable - a negative price would be tidy but wrong.

How is the source credited in the files?

Comment lines at the top of the CSV, a second sheet called Source in the Excel file, and a details section in the JSON.

Slide 9 - Who did what



Time	2:53 to 3:10 (about 17 seconds)
Who	Any member

Say

That's who built each part - every tab in the app is labelled with its member, and we built the app together.

Thank you. We're happy to take questions, and we can show any part of it running in the app.

Point to

- Each card as you name the member.

After the slides - the questions

The last 5 minutes are questions on what each member did. Each member's own document in docs/member_documents ends with 50 practice questions with answers - about 30 on their own part and 20 on the whole project.

- If a question is about someone else's part, it is fine to say 'Pasindu wrote that part - Pasindu, would you like to answer?'
- If you do not know, say so, and say what you would check. That is better than guessing.
- Offer to show it: 'We can show you that running in the app.' The live demo notes list what to show for each member.